COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE212 Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	10/29/2026	**DATE SUBMITTED:**	10/29/2026
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ACTIVITY NO. ___ Final Skills Exam

1. PROCEDURE OUTPUT

Apache for Enterprise service
Node Exporter
Prometheus for Monitoriing Tool
1. Create a repository and label it as "Final_Exam_Surname"

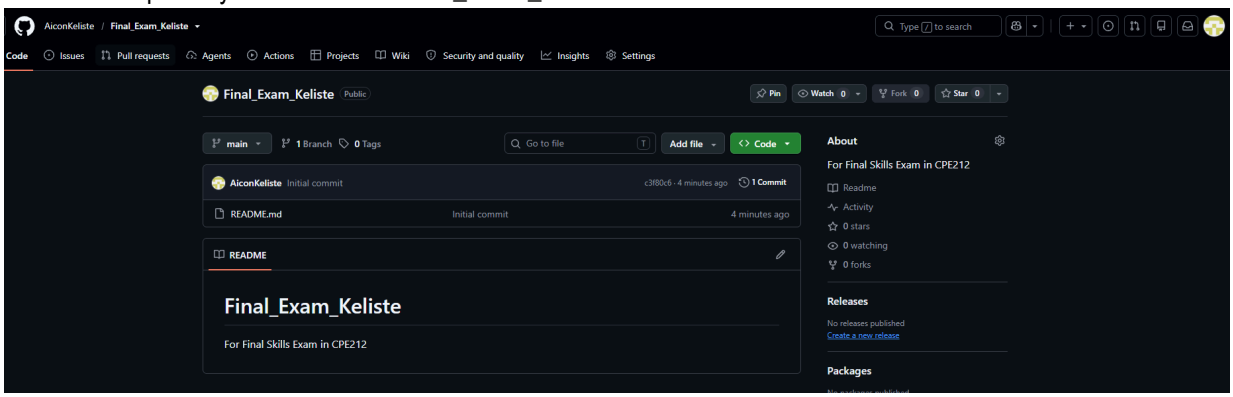


Figure 1. Created Repository

2. Clone your new repository in your VM

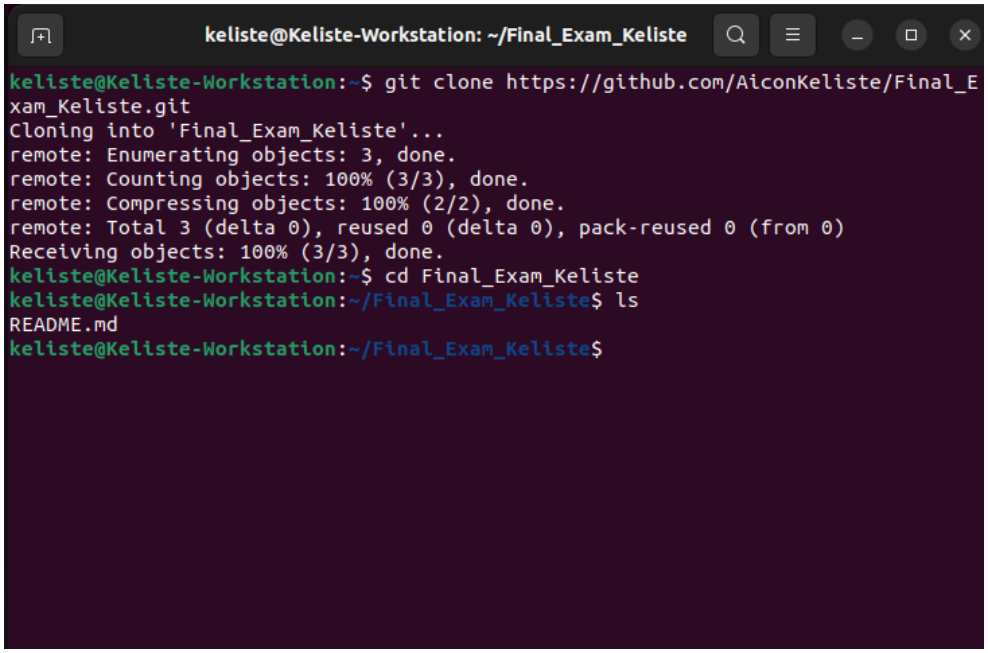


Figure 2 Github Repository Cloning

3. Create an Ansible playbook that does the following with an input of a config.yaml file and structure inventory file.

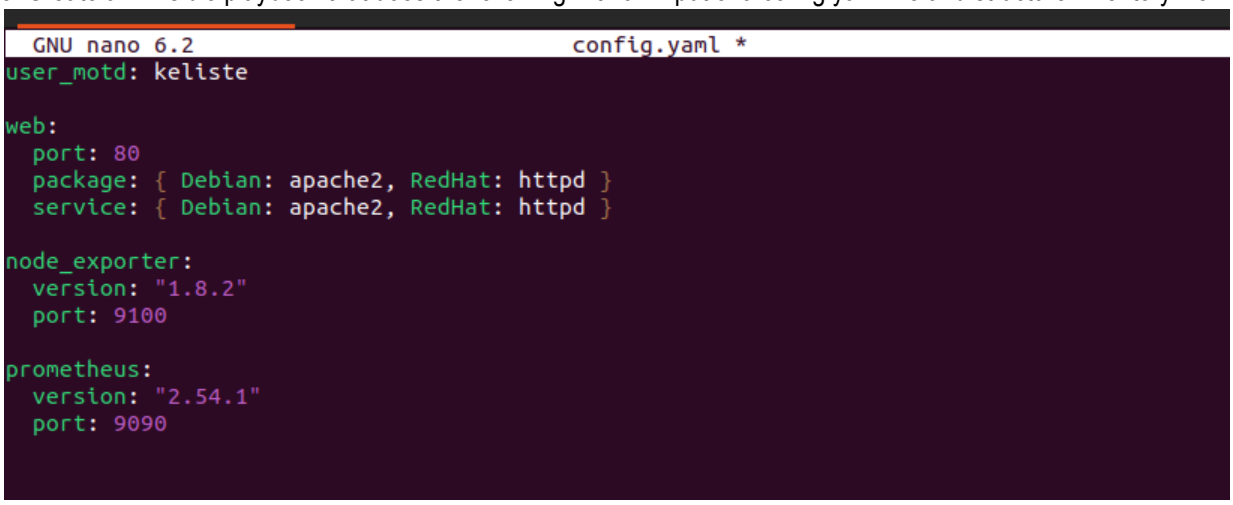


Figure 3 config.yaml

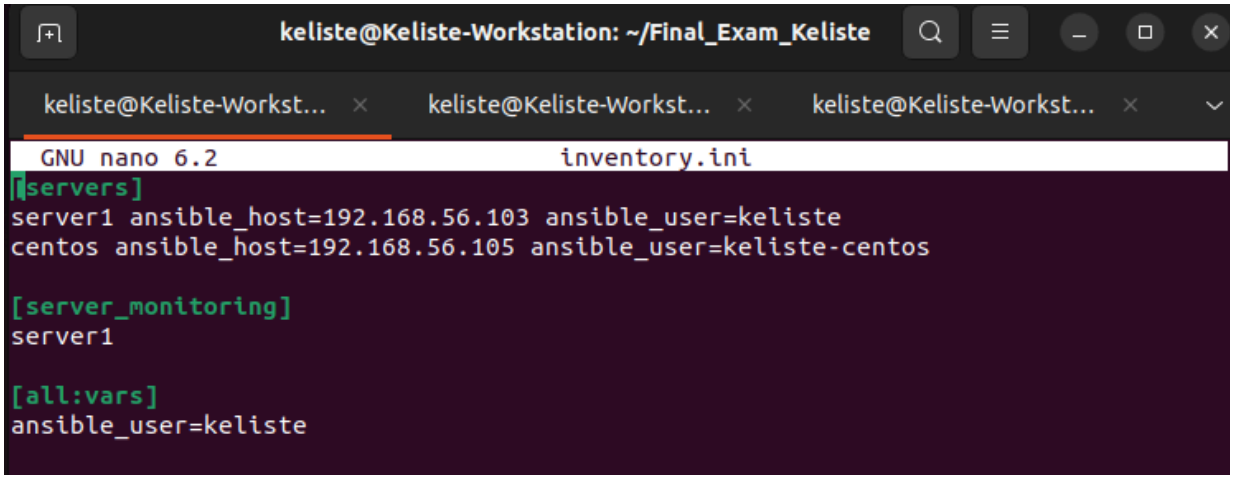


Figure 3.1 Inventory file

3.1 Install and configure one enterprise service that can be installed in Debian and Centos servers

```
PLAY [Apache, MOTD and node_exporter] ************************************

TASK [Gathering Facts] ***************************************************
ok: [server1]
[DEPRECATION WARNING]: Distribution centos 9 on host centos should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward
compatibility with prior Ansible releases. A future Ansible release will default
to using the discovered platform python for this host. See https://docs.ansible.c
om/ansible/2.10/reference_appendices/interpreter_discovery.html for more
information. This feature will be removed in version 2.12. Deprecation warnings
can be disabled by setting deprecation_warnings=False in ansible.cfg.
ok: [centos]

TASK [Update apt cache] **************************************************
skipping: [centos]
changed: [server1]

TASK [Install Apache] ****************************************************
changed: [server1]
ok: [centos]

TASK [Create index page] *************************************************
changed: [server1]
changed: [centos]

TASK [Start Apache] ******************************************************
ok: [server1]
ok: [centos]
```

Figure 3.3 Install and Start Apache Service

3.2 Install and configure one monitoring tool that can be installed in Debian and Centos servers (if it is a stack there should be option of different host)

```
TASK [Download and extract node_exporter] ***********************************************
fatal: [centos]: FAILED! => {"changed": false, "msg": "Failure downloading https://github.
om/prometheus/node_exporter/releases/download/v1.8.2/node_exporter-1.8.2.linux-amd64.tar.g
, The read operation timed out"}
ok: [server1]

TASK [Copy node_exporter binary] ********************************************************
ok: [server1]

TASK [Fix SELinux label on CentOS] ******************************************************
skipping: [server1]

TASK [Create node_exporter service] *****************************************************
ok: [server1]

TASK [Start node_exporter] **************************************************************
ok: [server1]
```

Figure 3.4 Node Exporter and Promethes Download and Install in Server 1 and Centos

4.4 Change Motd as "Ansible Managed by <username>"

```
    - name: Change MOTD
      ansible.builtin.copy:
        dest: /etc/motd
        content: "Ansible Managed by {{ user_motd }}\n"
```

Figure 3.5 Change the Motd

4. Push and commit your files in GitHub

```
keliste@Keliste-Workstation:~/Final_Exam_Keliste$ git add .
keliste@Keliste-Workstation:~/Final_Exam_Keliste$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        new file:   ansible.cfg
        new file:   config.yaml
        new file:   inventory.ini
        new file:   site.yml
        new file:   templates/prometheus.yml.j2

keliste@Keliste-Workstation:~/Final_Exam_Keliste$ git commit -m "Final SKills"
[main d177f3f] Final SKills
 5 files changed, 177 insertions(+)
 create mode 100644 ansible.cfg
 create mode 100644 config.yaml
 create mode 100644 inventory.ini
 create mode 100644 site.yml
```

```
keliste@Keliste-Workstation:~/Final_Exam_Keliste$ git push
Username for 'https://github.com': AiconKeliste
Password for 'https://AiconKeliste@github.com':
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Compressing objects: 100% (7/7), done.
Writing objects: 100% (8/8), 1.97 KiB | 1.97 MiB/s, done.
Total 8 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/AiconKeliste/Final_Exam_Keliste.git
   c3f80c6..d177f3f  main -> main
keliste@Keliste-Workstation:~/Final_Exam_Keliste$
```

5. Make sure to show evidence of input (codes) process (codes successfully running) and output (evidence of installation)

5. For your final exam to be counted, please paste your repository link as an answer in this exam.

Note: Extra points if you will implement the said services via containerization.

ssh keliste@192.168.56.103 "systemctl status prometheus --no-pager" cehcker